LAB 6

SETUP

- Installed the required dependencies for PyTorch training, audio I/O, visualization, perceptual/spectral losses, and CO2 tracking.
- Detected the available backend (CUDA or MPS or CPU) and set random seedsto make it global.
- Defined the main experiment params: 16 kHz sample rate and fixed 1 second clips (16,000 samples) to keep training consistent and efficient.

In [1]:
```python
import os, sys, zipfile, random, time, pathlib
import numpy as np
import torch
import torch.nn as nn
import torch.nn.functional as F
import torchaudio

try:
    import soundfile as sf
except Exception:
    !pip -q install soundfile
    import soundfile as sf

from torch.utils.data import Dataset, DataLoader
import pandas as pd
import matplotlib.pyplot as plt
from tqdm import trange
from pathlib import Path

try:
    from codecarbon import EmissionsTracker
    USE_CODECARBON = True
except Exception:
    !pip -q install codecarbon
    from codecarbon import EmissionsTracker
    USE_CODECARBON = True

CODECARBON_DIR = Path("./codecarbon_logs")
CODECARBON_DIR.mkdir(parents=True, exist_ok=True)

try:
    import psutil
    USE_PSUTIL = True
except Exception:
    !pip -q install psutil
    import psutil
    USE_PSUTIL = True

import subprocess

print("PWD:", pathlib.Path().resolve())
print("Root:", list(pathlib.Path("/mnt").glob("*")))

if torch.cuda.is_available():
    device = "cuda"
elif torch.backends.mps.is_available():
    device = "mps"
else:
    device = "cpu"
print("Device:", device)

torch.set_num_threads(max(1, (os.cpu_count() or 4) - 1))

SEED = 42
random.seed(SEED)
np.random.seed(SEED)
torch.manual_seed(SEED)
if device == "cuda":
    torch.cuda.manual_seed_all(SEED)

SAMPLE_RATE = 16000
TARGET_LENGTH = 16000
BATCH_SIZE = 64
N_EPOCHS = 50
N_EPOCHS_PERC = 50

DATA_ROOT = None

proc = psutil.Process(os.getpid()) if USE_PSUTIL else None

def _bytes_to_gb(x):
    return float(x) / (1024**3)

def get_cpu_ram():
    if proc is None:
        return None, None
    try:
        cpu = psutil.cpu_percent(interval=None)
        rss = _bytes_to_gb(proc.memory_info().rss)
        return float(cpu), float(rss)
    except Exception:
        return None, None

def get_gpu_mem():
    if device != "cuda":
        return None, None
    try:
        alloc = _bytes_to_gb(torch.cuda.memory_allocated())
        peak = _bytes_to_gb(torch.cuda.max_memory_allocated())
        return float(alloc), float(peak)
    except Exception:
        return None, None
```

```
[notice] A new release of pip is available: 24.3.1 -> 25.3
[notice] To update, run: pip install --upgrade pip

[notice] A new release of pip is available: 24.3.1 -> 25.3
[notice] To update, run: pip install --upgrade pip
PWD: /root
Root: [PosixPath('/mnt/mtat-datalabs')]
Device: cuda
```

DATA LOADING WITH MAGNATAGATUNE

Reutilizamos parcialmente el codigo del lab anterior

En lab 5 ya teniamos:

- Mirdata initialize
- Un dataset
- Segmentación

Implemented a robust loading strategy that works across environments:

- Modal Volume (if the dataset is mounted at /mnt/mtat-datalabs)
- mirdata (if the dataset is available in the installed mirdata version)
- Local fallback (searching for annotations_final.csv and clip_info_final.csv)

Summary: Built train/validation/test splits (80/10/10) and optionally capped the number of examples to control training time and cost (mainly when in need for it to execute fast)

In [2]:
```python
# intento de carga con mirdata y fallback a versión local o modal

USE_MIRDATA = False

# detectar Modal

IS_MODAL = bool(os.environ.get("MODAL_TASK_ID") or os.environ.get("MODAL_ENVIRONMENT") or os.environ.get("MODAL_RUN_ID"))
# ruta real modal
MODAL_VOLUME_ROOT = Path("/mnt/mtat-datalabs")

def read_csv_flexible(path):
    try:
        return pd.read_csv(path, sep="\t")
    except Exception:
        return pd.read_csv(path, sep=",")

def make_splits_from_dfs(df_ann, df_info, audio_root, max_tracks=10000):
    df_ann.columns = [c.strip() for c in df_ann.columns]
    df_info.columns = [c.strip() for c in df_info.columns]

    if "clip_id" not in df_ann.columns:
        raise ValueError(f"annotations_final.csv missing 'clip_id'. Columns: {list(df_ann.columns)[:20]}")

    if "clip_id" not in df_info.columns or "mp3_path" not in df_info.columns:
        raise ValueError(f"clip_info_final.csv must contain 'clip_id' and 'mp3_path'. Columns: {list(df_info.columns)[:20]}")

    all_ids = df_ann["clip_id"].values
    if len(all_ids) > max_tracks:
        all_ids = all_ids[:max_tracks]

    all_ids = list(all_ids)
    random.shuffle(all_ids)

    n_total = len(all_ids)
    n_train = int(0.8 * n_total)
    n_val = int(0.1 * n_total)

    train_ids = all_ids[:n_train]
    val_ids = all_ids[n_train:n_train + n_val]
    test_ids = all_ids[n_train + n_val:]

    df_info_indexed = df_info.set_index("clip_id")

    def make_df(id_list):
        rows = []
        for cid in id_list:
            if cid in df_info_indexed.index:
                rows.append({"clip_id": cid, "mp3_path": df_info_indexed.loc[cid, "mp3_path"]})
        return pd.DataFrame(rows)

    train_df = make_df(train_ids)
    val_df = make_df(val_ids)
    test_df = make_df(test_ids)

    return train_df, val_df, test_df, Path(audio_root)

if IS_MODAL and (MODAL_VOLUME_ROOT / "annotations_final.csv").exists():
    data_dir = str(MODAL_VOLUME_ROOT)
    ann_path = os.path.join(data_dir, "annotations_final.csv")
    info_path = os.path.join(data_dir, "clip_info_final.csv")

    # normalizar nombres de columnas por si vienen raros

    df_ann = read_csv_flexible(ann_path)
    df_info = read_csv_flexible(info_path)

    audio_root = os.path.join(data_dir, "audio")
    train_df, val_df, test_df, DATA_ROOT = make_splits_from_dfs(df_ann, df_info, audio_root)

    print("Using MagnaTagATune from Modal Volume.")
    print("Sizes (train, val, test):", len(train_df), len(val_df), len(test_df))
    print("DATA_ROOT:", DATA_ROOT)

else:
    try:
        import mirdata

        # algunos installs de mirdata no traen magnatagatune entonces si falla lo llevo al except

        data_home = os.environ.get("MTAT_MIRDATA_HOME", "./data/mtat")
        mtat = mirdata.initialize("magnatagatune", data_home=data_home)
        mtat.download()
        mtat.validate()

        all_tracks_dict = mtat.load_tracks()
        all_tracks = list(all_tracks_dict.values())
        random.shuffle(all_tracks)

        n_total = len(all_tracks)
        n_train = int(0.8 * n_total)
        n_val = int(0.1 * n_total)

        train_tracks = all_tracks[:n_train]
        val_tracks = all_tracks[n_train:n_train + n_val]
        test_tracks = all_tracks[n_train + n_val:]

        USE_MIRDATA = True
        print("Using MagnaTagATune via mirdata.")
        print("Sizes (train, val, test):", len(train_tracks), len(val_tracks), len(test_tracks))

    except Exception as e:
        print("mirdata not available or dataset not supported in this mirdata install. Falling back to local dataset.")
        print("Detail:", e)

        IN_COLAB = "google.colab" in sys.modules
        if IN_COLAB:
            from google.colab import drive
            drive.mount("/content/drive")
            search_roots = [Path("/content/drive/MyDrive"), Path("/content")]
        else:
            search_roots = [Path.cwd(), Path.cwd().parent, Path.cwd().parent.parent]

        data_path = None
        for root in search_roots:
            if root.exists():
                for p in root.rglob("annotations_final.csv"):
                    data_path = p.parent
                    break
            if data_path is not None:
                break

        if data_path is None:
            raise FileNotFoundError("Could not find annotations_final.csv in known locations")

        data_dir = str(data_path)
        ann_path = os.path.join(data_dir, "annotations_final.csv")
        info_path = os.path.join(data_dir, "clip_info_final.csv")

        df_ann = read_csv_flexible(ann_path)
        df_info = read_csv_flexible(info_path)

        audio_root = os.path.join(data_dir, "audio")
        os.makedirs(audio_root, exist_ok=True)

        full_zip = os.path.join(data_dir, "mp3_full.zip")
        if os.path.exists(full_zip) and not os.path.exists(os.path.join(audio_root, "000")):
            with zipfile.ZipFile(full_zip, "r") as z:
                z.extractall(audio_root)

        train_df, val_df, test_df, DATA_ROOT = make_splits_from_dfs(df_ann, df_info, audio_root)

        print("Using local MagnaTagATune.")
        print("Sizes (train, val, test):", len(train_df), len(val_df), len(test_df))
        print("DATA_ROOT:", DATA_ROOT)
```

```
mirdata not available or dataset not supported in this mirdata install. Falling back to local dataset.
Detail: No module named 'mirdata'
Using local MagnaTagATune.
Sizes (train, val, test): 8000 1000 1000
DATA_ROOT: /__modal/volumes/vo-IDPx0UuWAPUOrIsswHOuI/datalabs/audio
```

Dataset para encoder

AUTOENCODER BASELINE (MSE)

Autoencoder convolucional

Implemented a 1D convolutional autoencoder:

- Encoder with 3 Conv 1D layers (stride=2) to progressively downsample the waveform.
- Fixed size fully connected bottleneck of 512 (compression rate unchanged as required).
- Decoder with ConvTranspose1D layers to reconstruct the waveform back to the original length.

Trained a baseline model using MSE loss to obtain a quantitative reference and reconstruction baseline you can trust in time domain.

In [3]:
```python
# dataset para autoencoder (mirdata)
class MTATAutoencoderDatasetMirdata(Dataset):
    def __init__(self, tracks, split="train"):
        self.tracks = tracks
        self.split = split

    def __len__(self):
        return len(self.tracks)

    def __getitem__(self, idx):
        # carga de datos
        track = self.tracks[idx]
        audio, sr = track.audio

        # conversión a tensor
        audio = torch.tensor(audio, dtype=torch.float32)

        # conversión a mono
        if audio.ndim > 1:
            audio = audio.mean(dim=0)

        # resample
        if sr != SAMPLE_RATE:
            audio = torchaudio.functional.resample(
                audio.unsqueeze(0),
                orig_freq=sr,
                new_freq=SAMPLE_RATE
            ).squeeze(0)

        # tratamiento de audio largo (como seccion 2)
        audio = self._crop_or_pad(audio)

        return {"audio": audio}

    def _crop_or_pad(self, audio):
        length = audio.shape[-1]
        if length > TARGET_LENGTH:
            if self.split == "train":
                start = random.randint(0, length - TARGET_LENGTH)
            else:
                start = (length - TARGET_LENGTH) // 2
            audio = audio[start:start + TARGET_LENGTH]
        elif length < TARGET_LENGTH:
            pad = TARGET_LENGTH - length
            audio = F.pad(audio, (0, pad))
        return audio



# dataset para autoencoder (local: CSV + mp3)
class MTATAutoencoderDatasetLocal(Dataset):
    def __init__(self, df, split="train", data_root=None):
        self.df = df.reset_index(drop=True)
        self.split = split
        self.data_root = Path(data_root) if data_root is not None else None

    def __len__(self):
        return len(self.df)

    def __getitem__(self, idx):
        # carga de datos
        row = self.df.iloc[idx]
        mp3_rel = str(row["mp3_path"])

        # normalización de ruta (evita duplicar 'audio/audio/...')
        if mp3_rel.startswith("audio/"):
            mp3_rel = mp3_rel[len("audio/"):]

        audio_path = self.data_root / mp3_rel

        # lectura de audio (evita torchcodec/ffmpeg)
        # (T, C)
        audio_np, sr = sf.read(str(audio_path), always_2d=True)

        # conversión a tensor (C, T)
        audio = torch.tensor(audio_np, dtype=torch.float32).T

        # conversion a mono
        if audio.size(0) > 1:
            audio = audio.mean(dim=0, keepdim=True)

        # resample
        if sr != SAMPLE_RATE:
            audio = torchaudio.functional.resample(
                audio,
                orig_freq=sr,
                new_freq=SAMPLE_RATE
            )
            sr = SAMPLE_RATE

        audio = audio.squeeze(0)

        # tratamiento de audio largo (como seccion 2)
        audio = self._crop_or_pad(audio)

        return {"audio": audio}

    def _crop_or_pad(self, audio):
        length = audio.shape[-1]
        if length > TARGET_LENGTH:
            if self.split == "train":
                start = random.randint(0, length - TARGET_LENGTH)
            else:
                start = (length - TARGET_LENGTH) // 2
            audio = audio[start:start + TARGET_LENGTH]
        elif length < TARGET_LENGTH:
            pad = TARGET_LENGTH - length
            audio = F.pad(audio, (0, pad))
        return audio


# creación de datasets y dataloaders
if USE_MIRDATA:
    train_dataset = MTATAutoencoderDatasetMirdata(train_tracks, split="train")
    val_dataset   = MTATAutoencoderDatasetMirdata(val_tracks,   split="val")
    test_dataset  = MTATAutoencoderDatasetMirdata(test_tracks,  split="test")
else:
    if DATA_ROOT is None:
        raise ValueError("DATA_ROOT no está definido en fallback local")
    train_dataset = MTATAutoencoderDatasetLocal(train_df, split="train", data_root=DATA_ROOT)
    val_dataset   = MTATAutoencoderDatasetLocal(val_df,   split="valid", data_root=DATA_ROOT)
    test_dataset  = MTATAutoencoderDatasetLocal(test_df,  split="test",  data_root=DATA_ROOT)
import os


import platform

IS_MACOS = (platform.system().lower() == "darwin")

if IS_MACOS:
    NUM_WORKERS = 0
else:
    NUM_WORKERS = min(8, (os.cpu_count() or 8))

pin_memory = (device == "cuda")

train_loader = DataLoader(
    train_dataset,
    batch_size=BATCH_SIZE,
    shuffle=True,
    drop_last=True,
    num_workers=NUM_WORKERS,
    pin_memory=pin_memory,
    persistent_workers=(NUM_WORKERS > 0),
    prefetch_factor=2 if NUM_WORKERS > 0 else None
)

val_loader = DataLoader(
    val_dataset,
    batch_size=BATCH_SIZE,
    shuffle=False,
    drop_last=False,
    num_workers=NUM_WORKERS,
    pin_memory=pin_memory,
    persistent_workers=(NUM_WORKERS > 0),
    prefetch_factor=2 if NUM_WORKERS > 0 else None
)

test_loader = DataLoader(
    test_dataset,
    batch_size=BATCH_SIZE,
    shuffle=False,
    drop_last=False,
    num_workers=NUM_WORKERS,
    pin_memory=pin_memory,
    persistent_workers=(NUM_WORKERS > 0),
    prefetch_factor=2 if NUM_WORKERS > 0 else None
)

print("DataLoaders listos:")
print("Train batches:", len(train_loader))
print("Val batches:  ", len(val_loader))
print("Test batches: ", len(test_loader))
print("NUM_WORKERS:", NUM_WORKERS)

t0 = time.time()
_ = next(iter(train_loader))
print("Tiempo para 1 batch:", time.time() - t0, "sec")
```

```
DataLoaders listos:
Train batches: 125
Val batches:   16
Test batches:  16
NUM_WORKERS: 8
Tiempo para 1 batch: 3.0653979778289795 sec
```

In [4]:
```python
# perceptual / spectral loss (auraloss con fallback)
mse_loss = nn.MSELoss()

USE_AURALOSS = False
try:
    import auraloss
    mrstft = auraloss.freq.MultiResolutionSTFTLoss(
        fft_sizes=[512, 1024, 2048],
        hop_sizes=[128, 256, 512],
        win_lengths=[512, 1024, 2048]
    ).to(device)
    USE_AURALOSS = True

    def perceptual_loss(x_rec, x):
        return mrstft(x_rec, x)

except Exception:
    def multi_scale_stft_loss(x, y, scales=(512, 1024, 2048), hop_ratio=0.25, eps=1e-7):
        loss = 0.0
        for n_fft in scales:
            hop = int(n_fft * hop_ratio)
            window = torch.hann_window(n_fft, device=x.device)
            X = torch.stft(x.squeeze(1), n_fft=n_fft, hop_length=hop, window=window, return_complex=True)
            Y = torch.stft(y.squeeze(1), n_fft=n_fft, hop_length=hop, window=window, return_complex=True)
            Xmag = torch.log(X.abs() + eps)
            Ymag = torch.log(Y.abs() + eps)
            loss += torch.mean(torch.abs(Xmag - Ymag))
        return loss / len(scales)

    def perceptual_loss(x_rec, x):
        return multi_scale_stft_loss(x_rec, x)

def combined_loss(x_rec, x, alpha=0.01):
    l_mse = mse_loss(x_rec, x)
    l_stft = perceptual_loss(x_rec, x)
    return l_mse + alpha * l_stft, l_mse, l_stft

print("Perceptual loss:", "auraloss MRSTFT" if USE_AURALOSS else "manual multi-scale STFT")
```

```
Perceptual loss: manual multi-scale STFT
```

In [5]:
```python
# autoencoder convolucional
class AudioAutoencoder(nn.Module):
    def __init__(self, input_length=TARGET_LENGTH, bottleneck_size=512):
        super().__init__()
        self.input_length = input_length
        self.bottleneck_size = bottleneck_size

        # encoder
        self.enc_conv1 = nn.Conv1d(1, 16, kernel_size=9, stride=2, padding=4)
        self.enc_conv2 = nn.Conv1d(16, 32, kernel_size=9, stride=2, padding=4)
        self.enc_conv3 = nn.Conv1d(32, 64, kernel_size=9, stride=2, padding=4)

        length_after = input_length
        for _ in range(3):
            length_after = int(np.ceil(length_after / 2))

        self.flatten_dim = 64 * length_after

        # bottleneck (NO cambio compresión segun dice las instrucciones)
        self.fc_enc = nn.Linear(self.flatten_dim, bottleneck_size)
        self.fc_dec = nn.Linear(bottleneck_size, self.flatten_dim)

        # decoder
        self.dec_conv1 = nn.ConvTranspose1d(64, 32, kernel_size=9, stride=2, padding=4, output_padding=1)
        self.dec_conv2 = nn.ConvTranspose1d(32, 16, kernel_size=9, stride=2, padding=4, output_padding=1)
        self.dec_conv3 = nn.ConvTranspose1d(16, 1,  kernel_size=9, stride=2, padding=4, output_padding=1)

    def encode(self, x):
        x = F.relu(self.enc_conv1(x))
        x = F.relu(self.enc_conv2(x))
        x = F.relu(self.enc_conv3(x))
        x = x.flatten(start_dim=1)
        z = self.fc_enc(x)
        return z

    def decode(self, z):
        x = self.fc_dec(z)
        x = x.view(x.size(0), 64, -1)
        x = F.relu(self.dec_conv1(x))
        x = F.relu(self.dec_conv2(x))
        x = self.dec_conv3(x)
        x = x[..., :self.input_length]
        return x

    def forward(self, x):
        z = self.encode(x)
        x_rec = self.decode(z)
        return x_rec, z

#  check
tmp = AudioAutoencoder(input_length=TARGET_LENGTH, bottleneck_size=512).to(device)
n_params = sum(p.numel() for p in tmp.parameters() if p.requires_grad)
print("AudioAutoencoder listo. Params:", f"{n_params:,}")
del tmp
```

```
AudioAutoencoder listo. Params: 131,247,041
```

In [6]:
```python
from tqdm.auto import tqdm

def train_autoencoder(
    model,
    train_loader,
    val_loader,
    n_epochs=N_EPOCHS,
    lr=1e-3,
    use_perceptual=False,
    alpha=0.1,
    project_name="lab6_autoencoder"
):
    model = model.to(device)
    optimizer = torch.optim.Adam(model.parameters(), lr=lr)

    train_losses = []
    val_losses = []
    val_mse_list = []
    val_stft_list = []

    cpu_pct_per_epoch = []
    ram_gb_per_epoch = []
    gpu_alloc_gb_per_epoch = []
    gpu_peak_gb_per_epoch = []

    if device == "cuda":
        try:
            torch.cuda.reset_peak_memory_stats()
        except Exception:
            pass

    try:
        tracker = EmissionsTracker(project_name=project_name, log_level="error", save_to_file=True)
        tracker.start()
    except Exception:
        tracker = None

    t_global = time.time()

    pbar_epoch = tqdm(range(n_epochs), desc="Epochs", unit="epoch")
    for epoch in pbar_epoch:
        model.train()
        batch_train_losses = []

        pbar_train = tqdm(train_loader, desc=f"Train {epoch+1}/{n_epochs}", unit="batch", leave=False)
        for batch in pbar_train:
            x = batch["audio"].to(device, non_blocking=True).unsqueeze(1)

            optimizer.zero_grad(set_to_none=True)
            x_rec, _ = model(x)

            if use_perceptual:
                loss, _, _ = combined_loss(x_rec, x, alpha=alpha)
            else:
                loss = mse_loss(x_rec, x)

            loss.backward()
            optimizer.step()

            batch_train_losses.append(loss.item())
            pbar_train.set_postfix({"loss": f"{loss.item():.5f}"})

        epoch_train = float(np.mean(batch_train_losses))
        train_losses.append(epoch_train)

        model.eval()
        batch_val_losses = []
        batch_val_mse = []
        batch_val_stft = []

        with torch.no_grad():
            pbar_val = tqdm(val_loader, desc=f"Val {epoch+1}/{n_epochs}", unit="batch", leave=False)
            for batch in pbar_val:
                x = batch["audio"].to(device, non_blocking=True).unsqueeze(1)
                x_rec, _ = model(x)

                v_mse = mse_loss(x_rec, x)
                batch_val_mse.append(v_mse.item())

                if use_perceptual:
                    v_loss, _, v_stft = combined_loss(x_rec, x, alpha=alpha)
                    batch_val_losses.append(v_loss.item())
                    batch_val_stft.append(v_stft.item())
                    pbar_val.set_postfix({"stft": f"{v_stft.item():.5f}", "mse": f"{v_mse.item():.5f}"})
                else:
                    batch_val_losses.append(v_mse.item())
                    pbar_val.set_postfix({"mse": f"{v_mse.item():.5f}"})

        epoch_val = float(np.mean(batch_val_losses))
        epoch_val_mse = float(np.mean(batch_val_mse))
        val_losses.append(epoch_val)
        val_mse_list.append(epoch_val_mse)

        if len(batch_val_stft) > 0:
            val_stft_list.append(float(np.mean(batch_val_stft)))

        cpu_pct, ram_gb = get_cpu_ram()
        gpu_alloc_gb, gpu_peak_gb = get_gpu_mem()

        cpu_pct_per_epoch.append(cpu_pct)
        ram_gb_per_epoch.append(ram_gb)
        gpu_alloc_gb_per_epoch.append(gpu_alloc_gb)
        gpu_peak_gb_per_epoch.append(gpu_peak_gb)

        postfix = {
            "train": f"{epoch_train:.5f}",
            "val_mse": f"{epoch_val_mse:.5f}",
            "sec": f"{time.time()-t_global:.1f}"
        }
        if cpu_pct is not None:
            postfix["cpu%"] = f"{cpu_pct:.1f}"
        if ram_gb is not None:
            postfix["ramGB"] = f"{ram_gb:.2f}"
        if gpu_peak_gb is not None:
            postfix["gpuGB"] = f"{gpu_peak_gb:.2f}"

        pbar_epoch.set_postfix(postfix)

        print(f"[{epoch+1}/{n_epochs}] train={epoch_train:.6f} val_mse={epoch_val_mse:.6f}")

    emissions_kg = None
    if tracker is not None:
        try:
            emissions_kg = tracker.stop()
        except Exception:
            emissions_kg = None

    total_time_sec = time.time() - t_global

    def _safe_max(xs):
        xs2 = [x for x in xs if x is not None]
        return float(max(xs2)) if len(xs2) > 0 else None

    def _safe_mean(xs):
        xs2 = [x for x in xs if x is not None]
        return float(np.mean(xs2)) if len(xs2) > 0 else None

    return model, {
        "train_losses": train_losses,
        "val_losses": val_losses,
        "val_mse": val_mse_list,
        "val_stft": val_stft_list if len(val_stft_list) > 0 else None,
        "emissions_kg": emissions_kg,
        "time_sec": total_time_sec,
        "time_min": total_time_sec / 60.0,
        "cpu_pct_per_epoch": cpu_pct_per_epoch,
        "ram_gb_per_epoch": ram_gb_per_epoch,
        "gpu_alloc_gb_per_epoch": gpu_alloc_gb_per_epoch,
        "gpu_peak_gb_per_epoch": gpu_peak_gb_per_epoch,
        "cpu_pct_mean": _safe_mean(cpu_pct_per_epoch),
        "ram_gb_max": _safe_max(ram_gb_per_epoch),
        "gpu_peak_gb_max": _safe_max(gpu_peak_gb_per_epoch)
    }
```

In [7]:
```python
model_mse = AudioAutoencoder(input_length=TARGET_LENGTH, bottleneck_size=512).to(device)

model_mse, results_mse = train_autoencoder(
    model=model_mse,
    train_loader=train_loader,
    val_loader=val_loader,
    n_epochs=N_EPOCHS,
    lr=1e-3,
    use_perceptual=False,
    project_name="lab6_autoencoder_mse"
)

results_mse
```

```
[codecarbon WARNING @ 21:30:48] Multiple instances of codecarbon are allowed to run at the same time.
Epochs:    0%|          | 0/50 [00:00<?, ?epoch/s]
Train 1/50:    0%|          | 0/125 [00:00<?, ?batch/s]
Val 1/50:    0%|          | 0/16 [00:00<?, ?batch/s]
[1/50] train=0.027414 val_mse=0.026258
Train 2/50:    0%|          | 0/125 [00:00<?, ?batch/s]
Val 2/50:    0%|          | 0/16 [00:00<?, ?batch/s]
[2/50] train=0.025809 val_mse=0.026239
Train 3/50:    0%|          | 0/125 [00:00<?, ?batch/s]
Val 3/50:    0%|          | 0/16 [00:00<?, ?batch/s]
[3/50] train=0.025407 val_mse=0.026241
Train 4/50:    0%|          | 0/125 [00:00<?, ?batch/s]
Val 4/50:    0%|          | 0/16 [00:00<?, ?batch/s]
[4/50] train=0.025727 val_mse=0.026243
Train 5/50:    0%|          | 0/125 [00:00<?, ?batch/s]
Val 5/50:    0%|          | 0/16 [00:00<?, ?batch/s]
[5/50] train=0.025502 val_mse=0.026237
Train 6/50:    0%|          | 0/125 [00:00<?, ?batch/s]
Val 6/50:    0%|          | 0/16 [00:00<?, ?batch/s]
[6/50] train=0.025787 val_mse=0.026204
Train 7/50:    0%|          | 0/125 [00:00<?, ?batch/s]
Val 7/50:    0%|          | 0/16 [00:00<?, ?batch/s]
[7/50] train=0.025591 val_mse=0.026236
Train 8/50:    0%|          | 0/125 [00:00<?, ?batch/s]
Val 8/50:    0%|          | 0/16 [00:00<?, ?batch/s]
[8/50] train=0.025519 val_mse=0.025114
Train 9/50:    0%|          | 0/125 [00:00<?, ?batch/s]
Val 9/50:    0%|          | 0/16 [00:00<?, ?batch/s]
[9/50] train=0.020487 val_mse=0.018299
Train 10/50:    0%|          | 0/125 [00:00<?, ?batch/s]
Val 10/50:    0%|          | 0/16 [00:00<?, ?batch/s]
[10/50] train=0.016055 val_mse=0.015871
Train 11/50:    0%|          | 0/125 [00:00<?, ?batch/s]
Val 11/50:    0%|          | 0/16 [00:00<?, ?batch/s]
[11/50] train=0.014404 val_mse=0.014460
Train 12/50:    0%|          | 0/125 [00:00<?, ?batch/s]
Val 12/50:    0%|          | 0/16 [00:00<?, ?batch/s]
[12/50] train=0.013257 val_mse=0.013390
Train 13/50:    0%|          | 0/125 [00:00<?, ?batch/s]
Val 13/50:    0%|          | 0/16 [00:00<?, ?batch/s]
[13/50] train=0.012264 val_mse=0.012692
Train 14/50:    0%|          | 0/125 [00:00<?, ?batch/s]
Val 14/50:    0%|          | 0/16 [00:00<?, ?batch/s]
[14/50] train=0.011598 val_mse=0.011878
Train 15/50:    0%|          | 0/125 [00:00<?, ?batch/s]
Val 15/50:    0%|          | 0/16 [00:00<?, ?batch/s]
[15/50] train=0.010928 val_mse=0.011279
Train 16/50:    0%|          | 0/125 [00:00<?, ?batch/s]
Val 16/50:    0%|          | 0/16 [00:00<?, ?batch/s]
[16/50] train=0.010454 val_mse=0.010851
Train 17/50:    0%|          | 0/125 [00:00<?, ?batch/s]
Val 17/50:    0%|          | 0/16 [00:00<?, ?batch/s]
[17/50] train=0.010167 val_mse=0.010571
Train 18/50:    0%|          | 0/125 [00:00<?, ?batch/s]
Val 18/50:    0%|          | 0/16 [00:00<?, ?batch/s]
[18/50] train=0.009911 val_mse=0.010318
Train 19/50:    0%|          | 0/125 [00:00<?, ?batch/s]
Val 19/50:    0%|          | 0/16 [00:00<?, ?batch/s]
[19/50] train=0.009611 val_mse=0.010170
Train 20/50:    0%|          | 0/125 [00:00<?, ?batch/s]
Val 20/50:    0%|          | 0/16 [00:00<?, ?batch/s]
[20/50] train=0.009620 val_mse=0.010089
Train 21/50:    0%|          | 0/125 [00:00<?, ?batch/s]
Val 21/50:    0%|          | 0/16 [00:00<?, ?batch/s]
[21/50] train=0.009436 val_mse=0.010028
Train 22/50:    0%|          | 0/125 [00:00<?, ?batch/s]
Val 22/50:    0%|          | 0/16 [00:00<?, ?batch/s]
[22/50] train=0.009578 val_mse=0.010021
Train 23/50:    0%|          | 0/125 [00:00<?, ?batch/s]
Val 23/50:    0%|          | 0/16 [00:00<?, ?batch/s]
[23/50] train=0.009420 val_mse=0.009992
Train 24/50:    0%|          | 0/125 [00:00<?, ?batch/s]
Val 24/50:    0%|          | 0/16 [00:00<?, ?batch/s]
[24/50] train=0.009352 val_mse=0.009963
Train 25/50:    0%|          | 0/125 [00:00<?, ?batch/s]
Val 25/50:    0%|          | 0/16 [00:00<?, ?batch/s]
[25/50] train=0.009427 val_mse=0.010065
Train 26/50:    0%|          | 0/125 [00:00<?, ?batch/s]
Val 26/50:    0%|          | 0/16 [00:00<?, ?batch/s]
[26/50] train=0.009428 val_mse=0.009934
Train 27/50:    0%|          | 0/125 [00:00<?, ?batch/s]
Val 27/50:    0%|          | 0/16 [00:00<?, ?batch/s]
[27/50] train=0.009336 val_mse=0.009919
Train 28/50:    0%|          | 0/125 [00:00<?, ?batch/s]
```

```
Val 28/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[28/50] train=0.009516 val_mse=0.009904
Train 29/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 29/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[29/50] train=0.009348 val_mse=0.009891
Train 30/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 30/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[30/50] train=0.009307 val_mse=0.009891
Train 31/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 31/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[31/50] train=0.009375 val_mse=0.009863
Train 32/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 32/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[32/50] train=0.009446 val_mse=0.009867
Train 33/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 33/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[33/50] train=0.009377 val_mse=0.009855
Train 34/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 34/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[34/50] train=0.009256 val_mse=0.009903
Train 35/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 35/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[35/50] train=0.009242 val_mse=0.009832
Train 36/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 36/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[36/50] train=0.009331 val_mse=0.009810
Train 37/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 37/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[37/50] train=0.009234 val_mse=0.009820
Train 38/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 38/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[38/50] train=0.009395 val_mse=0.009839
Train 39/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 39/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[39/50] train=0.009348 val_mse=0.009827
Train 40/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 40/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[40/50] train=0.009167 val_mse=0.009823
Train 41/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 41/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[41/50] train=0.009385 val_mse=0.009796
Train 42/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 42/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[42/50] train=0.009326 val_mse=0.009775
Train 43/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 43/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[43/50] train=0.009085 val_mse=0.009761
Train 44/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 44/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[44/50] train=0.009331 val_mse=0.009752
Train 45/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 45/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[45/50] train=0.009212 val_mse=0.009779
Train 46/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 46/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[46/50] train=0.009256 val_mse=0.009749
Train 47/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 47/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[47/50] train=0.009272 val_mse=0.009826
Train 48/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 48/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[48/50] train=0.009190 val_mse=0.009734
Train 49/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 49/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[49/50] train=0.009232 val_mse=0.009752
Train 50/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 50/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[50/50] train=0.009312 val_mse=0.009721
```

```
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In [8]:

```
#N_EPOCHS = 8
```

In [9]:

```
# entrenamiento con perdida perceptual
model_perc = AudioAutoencoder(
    input_length=TARGET_LENGTH,
    bottleneck_size=512
).to(device)

model_perc, results_perc = train_autoencoder(
    model=model_perc,
    train_loader=train_loader,
    val_loader=val_loader,
    n_epochs=N_EPOCHS_PERC,
    lr=1e-3,
    use_perceptual=True,
    alpha=0.01,
    project_name="lab6_autoencoder_perc"
)

results_perc
```

```
Epochs:   0%|          | 0/50 [00:00<?, ?epoch/s]
Train 1/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 1/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[1/50] train=0.050706 val_mse=0.026619
Train 2/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 2/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[2/50] train=0.044708 val_mse=0.026494
Train 3/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 3/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[3/50] train=0.043845 val_mse=0.027060
Train 4/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 4/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[4/50] train=0.043638 val_mse=0.026823
Train 5/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 5/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[5/50] train=0.043060 val_mse=0.027000
Train 6/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 6/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[6/50] train=0.042756 val_mse=0.026626
Train 7/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 7/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[7/50] train=0.042555 val_mse=0.026753
Train 8/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 8/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[8/50] train=0.042244 val_mse=0.026661
Train 9/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 9/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[9/50] train=0.041956 val_mse=0.026593
Train 10/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 10/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[10/50] train=0.042068 val_mse=0.026682
Train 11/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 11/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[11/50] train=0.041753 val_mse=0.026700
Train 12/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 12/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[12/50] train=0.042299 val_mse=0.026821
Train 13/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 13/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[13/50] train=0.041581 val_mse=0.026505
Train 14/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 14/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[14/50] train=0.041806 val_mse=0.026674
Train 15/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 15/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[15/50] train=0.042043 val_mse=0.026864
Train 16/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 16/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[16/50] train=0.041875 val_mse=0.027651
Train 17/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 17/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[17/50] train=0.042101 val_mse=0.026973
Train 18/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 18/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[18/50] train=0.041969 val_mse=0.026744
Train 19/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 19/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[19/50] train=0.041201 val_mse=0.026810
Train 20/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 20/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[20/50] train=0.041033 val_mse=0.026523
Train 21/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 21/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[21/50] train=0.041513 val_mse=0.028340
Train 22/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 22/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[22/50] train=0.041429 val_mse=0.026664
Train 23/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 23/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[23/50] train=0.042223 val_mse=0.026825
Train 24/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 24/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[24/50] train=0.041166 val_mse=0.026594
Train 25/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 25/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[25/50] train=0.041003 val_mse=0.026712
Train 26/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 26/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[26/50] train=0.041549 val_mse=0.026624
Train 27/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 27/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[27/50] train=0.040939 val_mse=0.026434
Train 28/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 28/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[28/50] train=0.040763 val_mse=0.025821
Train 29/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 29/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[29/50] train=0.039205 val_mse=0.023346
Train 30/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 30/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[30/50] train=0.037577 val_mse=0.022550
[30/50] train=0.037577 val_mse=0.022550
Train 31/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 31/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[31/50] train=0.037880 val_mse=0.021507
Train 32/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 32/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[32/50] train=0.036194 val_mse=0.020276
Train 33/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 33/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[33/50] train=0.035117 val_mse=0.019518
Train 34/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 34/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[34/50] train=0.034388 val_mse=0.019013
Train 35/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 35/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[35/50] train=0.034516 val_mse=0.018965
Train 36/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 36/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[36/50] train=0.033473 val_mse=0.018119
Train 37/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 37/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[37/50] train=0.033241 val_mse=0.017969
Train 38/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 38/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[38/50] train=0.032187 val_mse=0.017483
Train 39/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 39/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[39/50] train=0.031923 val_mse=0.017027
Train 40/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 40/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[40/50] train=0.031817 val_mse=0.017625
Train 41/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 41/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[41/50] train=0.031650 val_mse=0.016750
Train 42/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 42/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[42/50] train=0.031288 val_mse=0.016787
Train 43/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 43/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[43/50] train=0.030800 val_mse=0.016208
Train 44/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 44/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[44/50] train=0.031079 val_mse=0.016432
Train 45/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 45/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[45/50] train=0.031016 val_mse=0.015349
Train 46/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 46/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[46/50] train=0.031936 val_mse=0.016239
Train 47/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 47/50:   0%|          | 0/16 [00:00<?, ?batch/s]
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Train 48/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 48/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[48/50] train=0.030401 val_mse=0.015707
Train 49/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 49/50:   0%|          | 0/16 [00:00<?, ?batch/s]
[49/50] train=0.029746 val_mse=0.015680
Train 50/50:   0%|          | 0/125 [00:00<?, ?batch/s]
Val 50/50:   0%|          | 0/16 [00:00<?, ?batch/s]
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```

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```
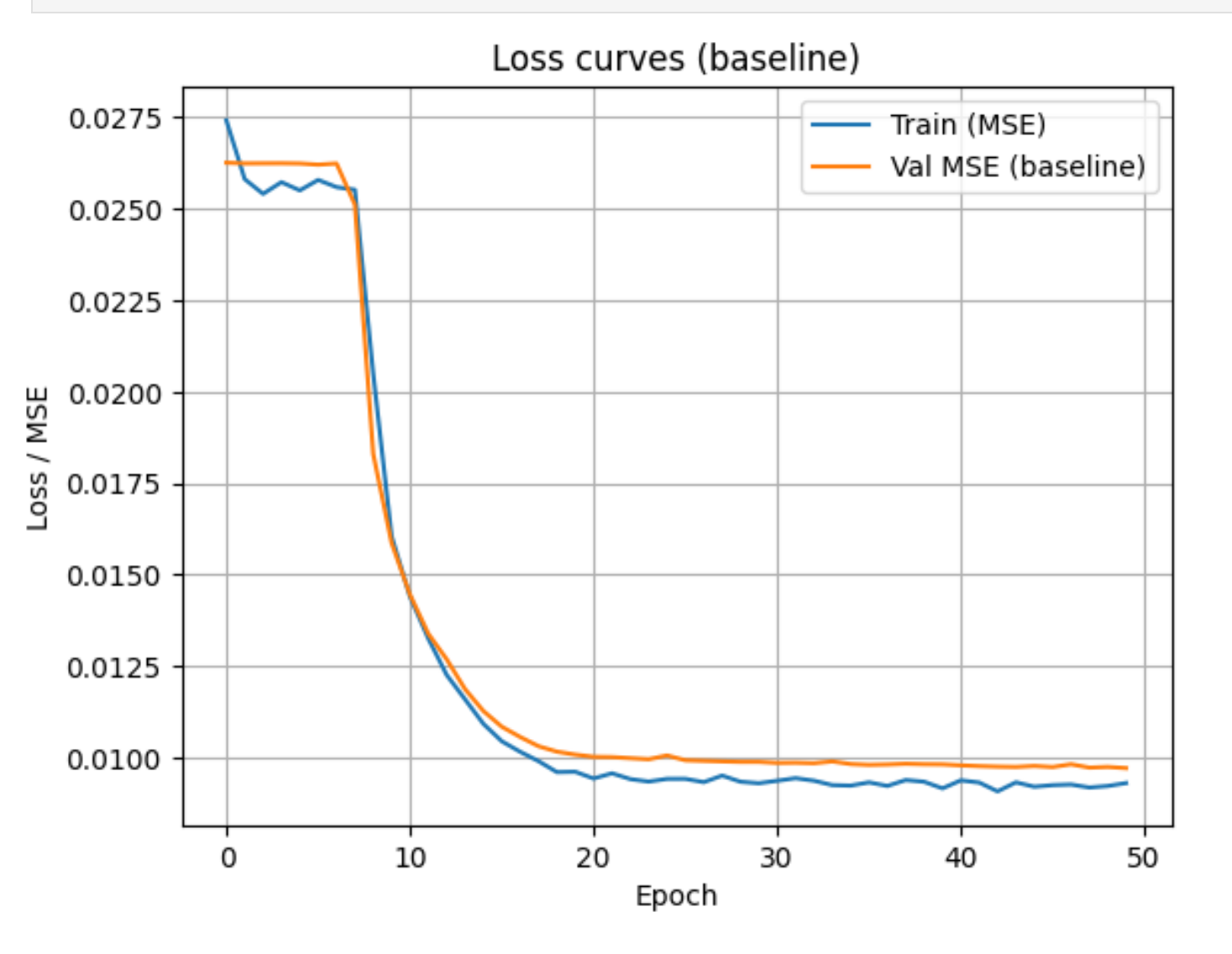
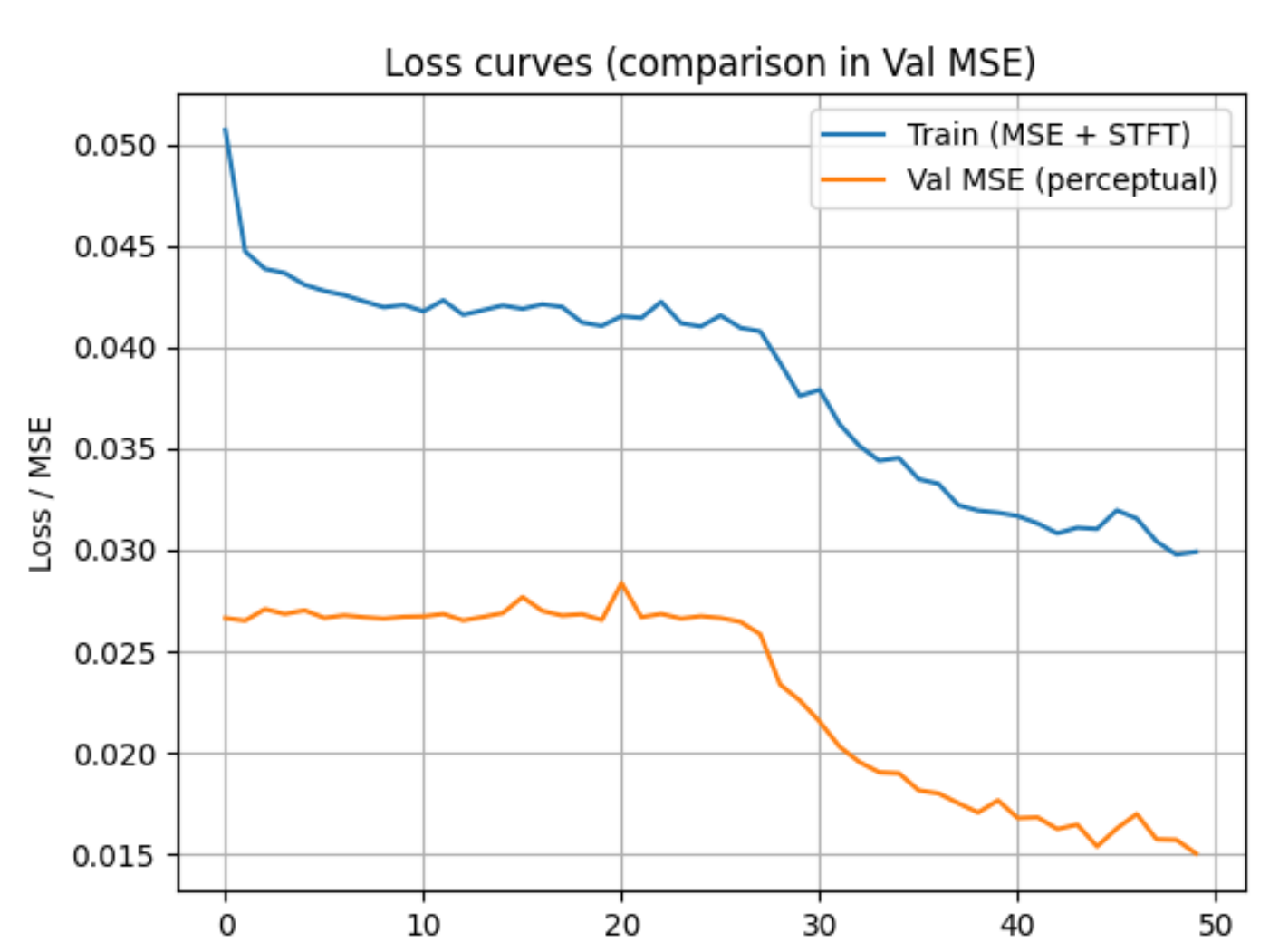
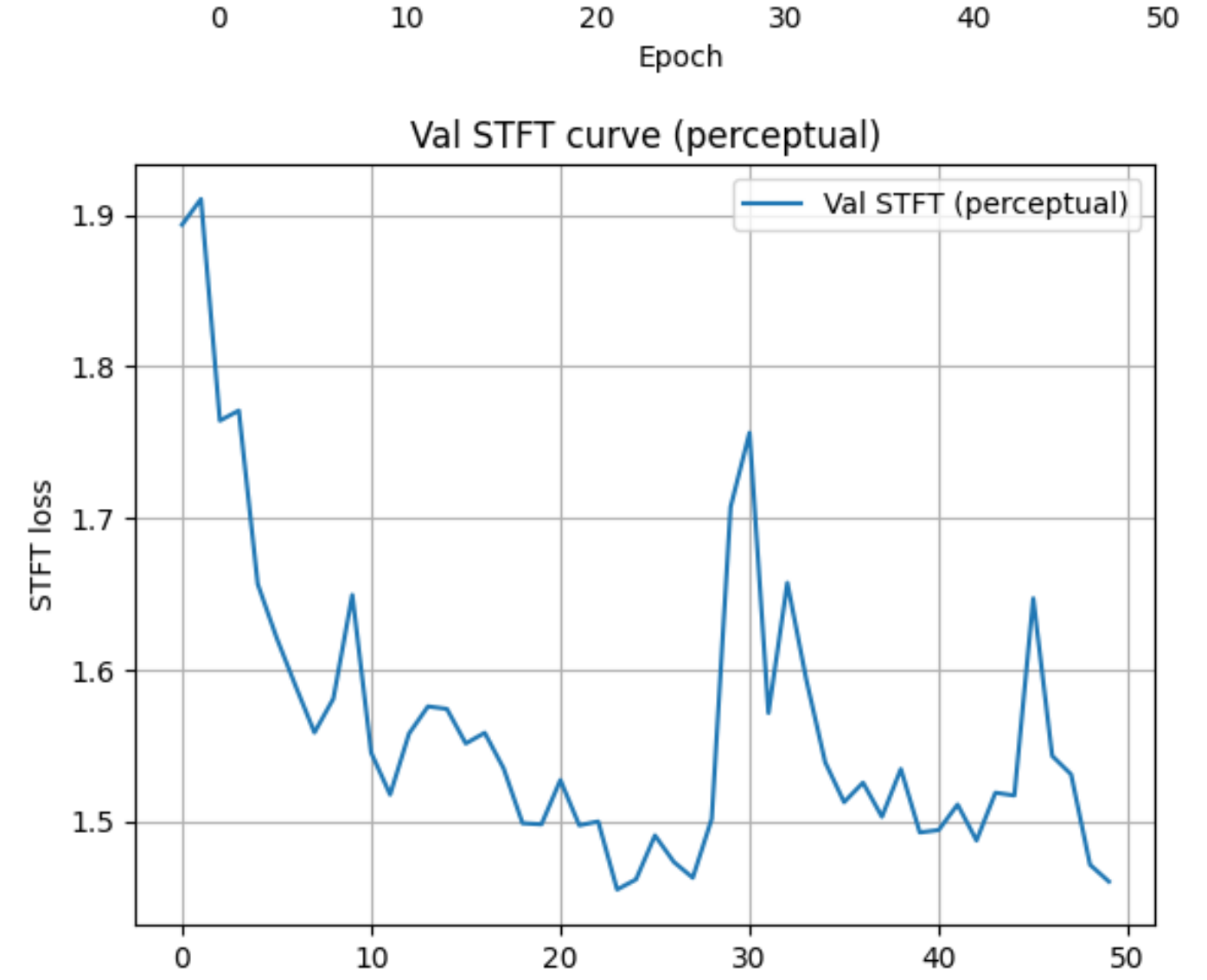

In [10]:

```python
# grafica de perdidas

plt.figure()
plt.plot(results_mse["train_losses"], label="Train (MSE)")
plt.plot(results_mse["val_mse"], label="Val MSE (baseline)")
plt.xlabel("Epoch")
plt.ylabel("Loss / MSE")
plt.legend()
plt.title("Loss curves (baseline)")
plt.grid(True)
plt.show()

if "results_perc" in globals():
    plt.figure()
    plt.plot(results_perc["train_losses"], label="Train (MSE + STFT)")
    plt.plot(results_perc["val_mse"], label="Val MSE (perceptual)")
    plt.xlabel("Epoch")
    plt.ylabel("Loss / MSE")
    plt.legend()
    plt.title("Loss curves (comparison in Val MSE)")
    plt.grid(True)
    plt.show()

    # grafica extra: solo para ver STFT val del perceptual

    if results_perc["val_stft"] is not None:
        plt.figure()
        plt.plot(results_perc["val_stft"], label="Val STFT (perceptual)")
        plt.xlabel("Epoch")
        plt.ylabel("STFT loss")
        plt.legend()
        plt.title("Val STFT curve (perceptual)")
        plt.grid(True)
        plt.show()
```

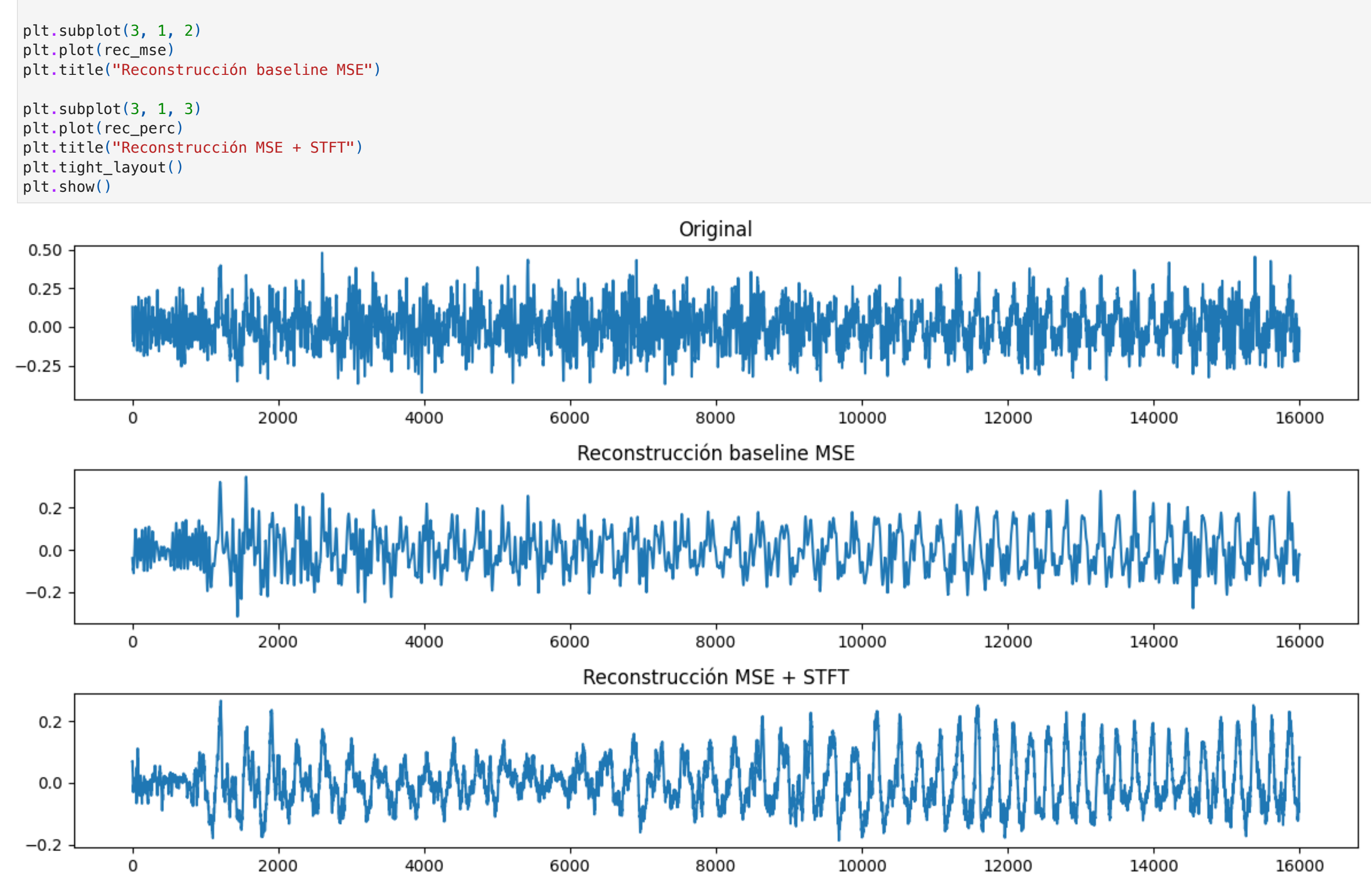






In [11]:

```python
# ejemplo de reconstruccion
from IPython.display import Audio, display

batch = next(iter(test_loader))
x = batch["audio"][:4].float().to(device).unsqueeze(1)

model_mse.eval()
model_perc.eval()

with torch.no_grad():
    x_rec_mse, _ = model_mse(x)
    x_rec_perc, _ = model_perc(x)

# waveform original vs reconstruida
idx = 0
orig = x[idx].detach().cpu().squeeze().numpy()
rec_mse = x_rec_mse[idx].detach().cpu().squeeze().numpy()
rec_perc = x_rec_perc[idx].detach().cpu().squeeze().numpy()

plt.figure(figsize=(12, 6))
plt.subplot(3, 1, 1)
plt.plot(orig)
plt.title("Original")

plt.subplot(3, 1, 2)
plt.plot(rec_mse)
plt.title("Reconstrucción baseline MSE")

plt.subplot(3, 1, 3)
plt.plot(rec_perc)
plt.title("Reconstrucción MSE + STFT")
plt.tight_layout()
plt.show()
```



In [12]:

```python
from IPython.display import Audio, display
import numpy as np
import torch

def rms(x, eps=1e-8):
    return np.sqrt(np.mean(x**2) + eps)

def match_rms(y, ref, eps=1e-8):
    ry = rms(y, eps)
    rr = rms(ref, eps)
    return y * (rr / (ry + eps))

def to_audio(x_t):
    return x_t.detach().cpu().squeeze().numpy()

def show_examples(model_a, model_b, loader, n=12, seed=0):
    g = torch.Generator()
    g.manual_seed(seed)

    model_a.eval()
    model_b.eval()

    batch = next(iter(loader))
    x = batch["audio"].float()

    if x.shape[0] < n:
        n = x.shape[0]

    idxs = torch.randperm(x.shape[0], generator=g)[:n].tolist()
    x_sel = x[idxs].to(device).unsqueeze(1)

    with torch.no_grad():
        ra, _ = model_a(x_sel)
        rb, _ = model_b(x_sel)

    for i in range(n):
        orig = to_audio(x_sel[i])
        a = to_audio(ra[i])
        b = to_audio(rb[i])

        a = match_rms(a, orig)
        b = match_rms(b, orig)

        a = np.clip(a, -1.0, 1.0)
        b = np.clip(b, -1.0, 1.0)

        print(f"Example {i+1}/{n}")
        print("Original:")
        display(Audio(orig, rate=SAMPLE_RATE))
        print("Recon (MSE):")
        display(Audio(a, rate=SAMPLE_RATE))
        print("Recon (Perceptual):")
        display(Audio(b, rate=SAMPLE_RATE))
        print("-" * 60)

show_examples(model_mse, model_perc, test_loader, n=12, seed=42)
```

```
Example 1/12
Original:
▶ 0:00 -0:01

Recon (MSE):
▶ 0:00 -0:01

Recon (Perceptual):
▶ 0:00 -0:01

------------------------------------------------------------
Example 2/12
Original:
▶ 0:00 -0:01

Recon (MSE):
▶ 0:00 -0:01

Recon (Perceptual):
▶ 0:00 -0:01

------------------------------------------------------------
Example 3/12
Original:
▶ 0:00 -0:01

Recon (MSE):
▶ 0:00 -0:01

Recon (Perceptual):
▶ 0:00 -0:01

------------------------------------------------------------
Example 4/12
Original:
▶ 0:00 -0:01

Recon (MSE):
▶ 0:00 -0:01

Recon (Perceptual):
▶ 0:00 -0:01

------------------------------------------------------------
Example 5/12
Original:
▶ 0:00 -0:01

Recon (MSE):
▶ 0:00 -0:01

Recon (Perceptual):
▶ 0:00 -0:01

------------------------------------------------------------
Example 6/12
Original:
▶ 0:00 -0:01

Recon (MSE):
▶ 0:00 -0:01

Recon (Perceptual):
▶ 0:00 -0:01

------------------------------------------------------------
Example 7/12
Original:
▶ 0:00 -0:01

Recon (MSE):
▶ 0:00 -0:01

Recon (Perceptual):
▶ 0:00 -0:01

------------------------------------------------------------
Example 8/12
Original:
▶ 0:00 -0:01

Recon (MSE):
▶ 0:00 -0:01

Recon (Perceptual):
▶ 0:00 -0:01

------------------------------------------------------------
Example 9/12
Original:
▶ 0:00 -0:01

Recon (MSE):
▶ 0:00 -0:01

Recon (Perceptual):
▶ 0:00 -0:01

------------------------------------------------------------
Example 10/12
Original:
▶ 0:00 -0:01

Recon (MSE):
▶ 0:00 -0:01

Recon (Perceptual):
▶ 0:00 -0:01

------------------------------------------------------------
Example 11/12
Original:
▶ 0:00 -0:01

Recon (MSE):
▶ 0:00 -0:01

Recon (Perceptual):
▶ 0:00 -0:01

------------------------------------------------------------
Example 12/12
Original:
▶ 0:00 -0:01

Recon (MSE):
▶ 0:00 -0:01

Recon (Perceptual):
▶ 0:00 -0:01

------------------------------------------------------------
```

LONG AUDIOS STRATEGY

Handled variable and length audio by enforcing a fixed input length (1 second):

- Training: random crop (seeking to add temporal augmentation and improve generalization).
- Validation/Test: centered crop (to ensure consistent evaluation).

This strategy is expected to allow training the model on MTAT clips with different durations while keeping batch shapes fixed.

RECONSTRUCTION IMPROVEMENT (PERCEPTUAL LOSS)

- Trained a second model using a perceptual loss based on multi resolution STFT trying to combine it with MSE.* The goal is to improve perceived spectral quality of reconstructions (timbre, detail), even if there's not a notable improvement in MSE.

RESULTS AND REPORT

- Reported training and validation curves and compared both models using a common metric (Validation MSE).
- For the perceptual model, also tracked the validation perceptual/STFT loss, mainly as an auxiliary metric.
- Measured total training time and estimated CO2 emissions using CodeCarbon for resource (and) environmental reporting. (Although when running in)

In [13]:

```python
# resumen final de resultados y emisiones
def print_summary(name, res):
    print(f"=== {name} ===")

    # métrica comparable entre modelos
    if "val_mse" in res and res["val_mse"] is not None:
        print(f"Best Val MSE:  {min(res['val_mse']):.6f}")

    # métrica perceptual (solo aplica si existe)
    if "val_stft" in res and res["val_stft"] is not None:
        print(f"Best Val STFT: {min(res['val_stft']):.6f}")

    print(f"Total time:  {res['time_min']:.2f} min")
    print(f"Total CO2 Emissions: {res['emissions_kg']:.6f} kg")
    print()

print_summary("Baseline MSE", results_mse)
print_summary("MSE + STFT",   results_perc)


print("CPU mean (%):", results_mse.get("cpu_pct_mean"), "| RAM max (GB):", results_mse.get("ram_gb_max"), "| GPU peak (GB):", results_mse.get("gpu_peak_
print("CPU mean (%):", results_perc.get("cpu_pct_mean"), "| RAM max (GB):", results_perc.get("ram_gb_max"), "| GPU peak (GB):", results_perc.get("gpu_pea
```

```
=== Baseline MSE ===
Best Val MSE:  0.009721
Total time:  35.15 min
Total CO2 Emissions: 0.045078 kg

=== MSE + STFT ===
Best Val MSE:  0.015000
Best Val STFT: 1.455055
Total time:  40.66 min
Total CO2 Emissions: 0.052221 kg

CPU mean (%): 0.0 | RAM max (GB): 6.573619842529297 | GPU peak (GB): 2.4769086837768555
CPU mean (%): 0.0 | RAM max (GB): 6.8452301025390625 | GPU peak (GB): 3.449263572692871
```